
ReBel: Rescuing Step-Level Credit Assignment via Hierarchical Belief-Observation Grouping for LLM Agent Training

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Abstract

Reinforcement learning for interactive LLM agents demands accurate step-level credit assignment, yet existing step-level methods such as GiGPO suffer from a previously uncharacterised failure mode: 63–85% of all training steps form *singleton groups* under exact observation hashing, yielding a step-level advantage of zero and wasting the majority of the available learning signal. We present **ReBel** (Reinforcement Learning with Belief-State Enhancement), a training framework that addresses this limitation by making structured belief output the unifying primitive across three mutually reinforcing components. **Structured belief prompting** furnishes each step with an explicit JSON world model that simultaneously provides a richer grouping signal, a basis for intrinsic reward, and a cognitive scaffold—properties that only manifest when a step-level credit mechanism is present to exploit them. A **competence-adaptive belief reward curriculum** delivers dense intrinsic feedback during early training while differentially suppressing reward components that conflict with the task objective (Exploration decays $2.5\times$ faster than Progress), directly addressing the reward-shaping bias that makes undecayed dense belief reward harmful (−3.9 pp in ablation). Targeted ablation under the full ReBel system confirms this curriculum is the dominant performance contributor: removing it costs −3.1 pp peak SR (Table 4); replacing its adaptive schedule with cosine decay costs a further −3.1 pp, reflecting the qualitative difference in how quickly misaligned components are suppressed. **HiBO** (Hierarchical Belief-Observation grouping) applies these belief states as a semantic fallback for singleton steps, raising effective step-advantage coverage from $\sim 20\%$ to $\sim 76\%$ and contributing +0.8 pp peak SR in targeted ablation; its primary measured value is improved training stability rather than isolated peak performance. On ALFWorld, the full ReBel system achieves 95.3% peak success rate (seed = 42), surpassing GRPO by 13.3 pp and the published GiGPO by 3.9 pp. On WebShop, ReBel reaches 74.0% mean peak SR across three seeds as a preliminary transfer result; baseline comparisons are in preparation.

1 Introduction

Reinforcement learning (RL) for large language model (LLM) agents has emerged as a compelling paradigm for teaching models to solve multi-step interactive tasks: household manipulation [Shridhar et al., 2021], web navigation [Yao et al., 2022], and scientific reasoning. In these settings, an agent must issue a sequence of decisions over dozens of steps, where success depends not merely on individual actions but on their coherent composition into task-completing strategies.

The step-level credit assignment problem. A fundamental challenge in this regime is *credit assignment*: which of the many actions in a trajectory contributed to a successful outcome? Episode-level methods such as GRPO [Shao et al., 2024] assign the same reward signal to every step in a

trajectory, ignoring the heterogeneous quality of individual decisions. Step-level methods, most prominently GiGPO [Feng et al., 2025], improve on this by grouping trajectories that share the same environment state and comparing the downstream outcomes of different actions at that state—a form of counterfactual credit assignment.

The singleton group problem. Despite its theoretical appeal, step-level grouping in practice suffers from a critical failure mode that has not previously been identified and addressed. GiGPO groups trajectories by *exact observation text hash*: two steps belong to the same group only if their observation strings are character-for-character identical. In interactive text environments such as ALFWorld, observations enumerate objects in non-deterministic order and vary in wording; even two trajectories that reach logically identical states will typically produce different observation strings. As a result, **63–85% of all training steps form singleton groups**, for which the step-level advantage is mathematically zero (the group mean equals the single sample value). GiGPO’s step mechanism is therefore active for only 15–30% of steps at any given point in training. The paradox is that this sparse signal still delivers a substantial gain over GRPO (+6.2% in our controlled ablation), which implies that the remaining 70–85% of discarded steps carry proportionally valuable learning signal.

Our approach: ReBel. We propose to *rescue* this discarded signal through a principled use of belief states. At each step, an LLM agent can produce a structured belief state—a compact summary of what it knows about the world, its current subgoal, and its exploration history. Two steps that share the same observation might not hash identically, but they can share a belief state cluster (e.g., “holding knife, searching for fridge”), providing a semantically justified basis for grouping.

ReBel exploits this observation through three complementary mechanisms (Figure 2):

1. **Structured belief prompting:** a JSON-formatted belief template that forces the agent to articulate its world model, subgoal status, and exploration map at every step. This is the practical foundation for the other components: reliable belief extraction enables HiBO’s semantic fallback grouping and grounds the intrinsic reward components; without parseable belief output, both mechanisms fall back to their simpler equivalents. Empirically, switching from `<think>` to `<belief>` format *within GiGPO* already accounts for +3.1 pp peak SR (91.4% → 94.5%; Table 1), establishing that richer intermediate representation substantially amplifies the step-level credit signal even before algorithmic changes.
2. **Competence-adaptive belief reward curriculum:** dense intrinsic rewards computed from the agent’s belief quality, weighted by a curriculum that decays adaptively as the agent improves. Within the full ReBel system, targeted ablation (Table 4) shows removing the belief reward costs −3.1 pp peak SR, and replacing the adaptive schedule with cosine decay costs an additional −3.1 pp—together making the curriculum the dominant performance contributor at the performance frontier where ReBel operates. Different reward components decay at different rates: the exploration reward (which conflicts with depth-first task requirements) decays $2.5\times$ faster than the progress reward, mitigating the reward-shaping bias identified by Ng et al. [1999].
3. **HiBO (Hierarchical Belief-Observation Grouping):** a two-layer grouping strategy that preserves GiGPO’s high-quality exact observation groups where they exist and falls back to coarse-grained semantic belief clustering for singleton steps, raising coverage from $\sim 20\%$ to $\sim 76\%$. Within the full ReBel system, HiBO contributes +0.8 pp peak SR (Table 4); its primary measured value is the mechanistic validation that step-coverage can be substantially expanded via belief abstraction, and improved training stability in late epochs. We note that peak SR alone understates or overstates HiBO’s role depending on checkpoint selection; multi-seed experiments would provide a more reliable estimate of its variance-reducing contribution.

The three components form a closed loop: structured belief output provides the signal for HiBO and reward computation; the reward curriculum improves belief quality during early training; improved belief quality in turn enhances HiBO’s grouping reliability.

Contributions.

- We identify and quantify the singleton group problem in GiGPO-style step-level credit assignment (Section 3), demonstrating that 63–85% of step-level learning signal is routinely discarded.

- We show that switching from `<think>` to structured `<belief>` format within GiGPO already yields +3.1 pp peak SR (91.4% $\rightarrow$ 94.5%), establishing that richer intermediate representation substantially amplifies the value of step-level credit assignment (Table 1).
- We introduce a **competence-adaptive belief reward curriculum** with differential component decay; targeted ablation confirms it is the dominant performance contributor in the full system (−3.1 pp without belief reward; −3.1 pp with cosine vs. adaptive decay; Table 4), while also providing the mechanism by which dense belief reward avoids the harmful shaping bias seen without curriculum.
- We propose **HiBO**, a hierarchical grouping algorithm that raises step-advantage coverage from $\sim 20\%$ to $\sim 76\%$ via semantic belief abstraction (Section 4.1); targeted ablation shows +0.8 pp peak SR within the full system, with its primary measured value being training stability at the performance frontier.
- The three components *jointly* achieve 95.3% peak SR on ALFWorld (+13.3 pp over GRPO, +3.9 pp over published GiGPO, seed = 42) and 74.0% mean peak SR on WebShop across three seeds as a preliminary transfer result (Section 5.9).

2 Related Work

RL for LLM agents. Applying RL to train LLM-based agents has gained traction following the success of RLHF [Ouyang et al., 2022]. GRPO [Shao et al., 2024] and RLOO [Ahmadian et al., 2024] extend group-relative policy optimization to agentic settings, treating each trajectory as a sample for advantage estimation. RAGEN [Wang et al., 2025] studies multi-turn RL for agent training. GiGPO [Feng et al., 2025] introduces step-level credit assignment through observation-based trajectory grouping, achieving state-of-the-art results on ALFWorld. Our work builds on GiGPO’s step-level insight but addresses the fundamental singleton group limitation that renders it ineffective for the majority of steps.

Prompting and reasoning for agents. ReAct [Yao et al., 2023] pioneers the combination of reasoning traces and actions. Reflexion [Shinn et al., 2024] adds verbal self-reflection after failures. These methods rely on inference-time prompting without updating model parameters. ReBel’s structured belief prompting is distinct: it shapes the model’s output format to enable step-level grouping and reward computation during RL training.

Belief states and world models. The idea of maintaining an explicit belief state dates to model-based RL [Sutton, 1991] and partially observable Markov decision processes. In the LLM setting, belief prompting has been explored as a cognitive scaffold to improve agent consistency. ReBel is, to our knowledge, the first to leverage structured belief output simultaneously as a grouping signal and a reward basis within RL training for text-based agent benchmarks, creating a functional loop between belief maintenance, credit assignment, and reward shaping.

Reward shaping. Ng et al. [1999] establish that potential-based reward shaping preserves the optimal policy, while non-potential-based shaping introduces bias proportional to the shaping weight. Belief-based intrinsic rewards are not potential-based (they depend on model outputs), so they introduce policy bias without careful management. Our differential decay curriculum addresses this by driving the shaping coefficient to zero as the agent converges, with component-specific rates calibrated to each reward’s alignment with the task objective.

3 Background: GiGPO and the Singleton Group Problem

3.1 GiGPO Step-Level Advantage Estimation

GiGPO [Feng et al., 2025] decomposes the PPO advantage into episode and step components:

$$A_{\text{total}}(i) = A_{\text{episode}}(i) + \lambda \cdot A_{\text{step}}(i), \quad (1)$$

where A_{episode} is computed by GRPO-style group normalization across all rollouts sharing the same prompt, and A_{step} groups trajectories by the exact observation text at each step:

$$A_{\text{step}}(i) = G_t(i) - \text{mean}_{j \in \mathcal{G}(i)} G_t(j), \quad (2)$$

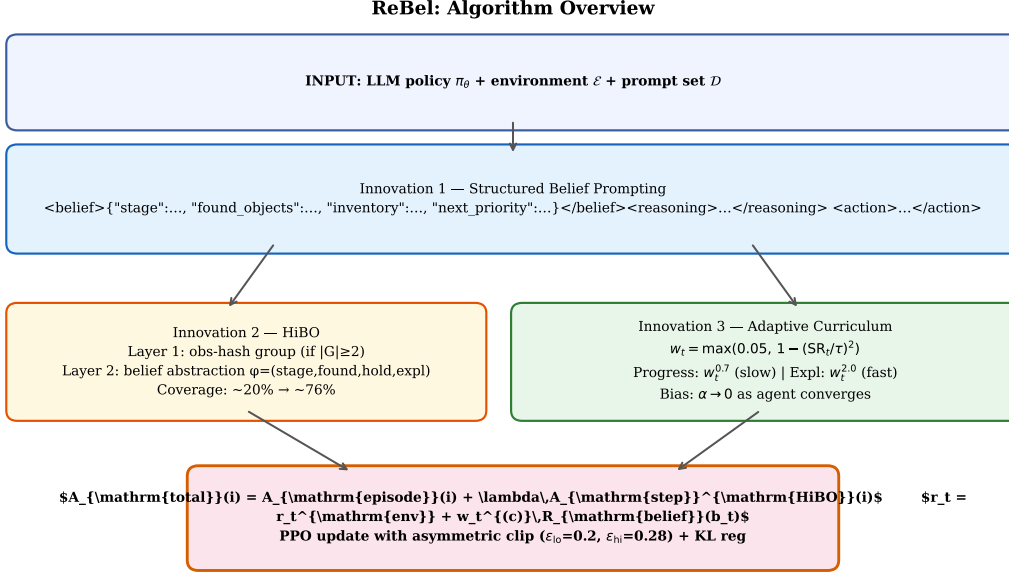


Figure 2: **ReBel overview.** Three innovations work in a closed loop: structured belief prompting produces JSON-formatted belief states that serve as both the grouping signal for HiBO and the basis for the intrinsic reward curriculum. HiBO recovers step-level learning signal from singleton observation groups. The adaptive curriculum decays differentially across reward components as the agent’s competence grows.

where $G_t(i)$ is the discounted return from step t in trajectory i , and $\mathcal{G}(i)$ is the observation group containing step i .

The counterfactual semantics of A_{step} are compelling: it answers “given *exactly* this environment state, how did my action compare to the average action taken here?” However, this semantics requires at least two trajectories to reach *identical* observations—and in practice they rarely do.

3.2 The Singleton Group Problem

ALFWorld observations enumerate room objects in non-deterministic order and use variable descriptions. In a typical batch of $n = 16$ rollouts per prompt, each covering up to 30 steps, the vast majority of step-level observations are unique. Table 5 (Section 5.4) quantifies this: 63–85% of steps form singleton groups, with median group size 1.0 throughout training. For such steps, Eq. (2) yields $A_{\text{step}} = G_t - G_t = 0$.

The step-level mechanism operates on only 15–30% of training data despite covering 100% of parameters. Yet this sparse signal still beats episode-level GRPO by 6.2 percentage points in our controlled ablation, indicating that the discarded 70–85% of steps carries proportionally valuable, recoverable information.

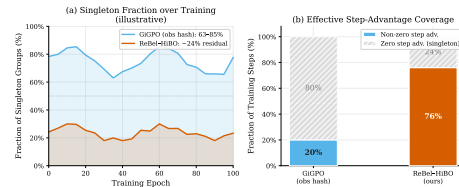


Figure 1: Singleton group prevalence in GiGPO. Over 63–85% of all steps form groups of size 1, yielding $A_{\text{step}} = 0$ for those steps.

4 Method: ReBel

4.1 Innovation 1: Hierarchical Belief-Observation Grouping (HiBO)

HiBO applies a two-layer precedence rule to determine which group each step belongs to for step-level advantage computation.

Layer 1 — Exact observation grouping. For each step i with observation o_i and prompt identifier u_i , we compute $\text{key}_1(i) = (u_i, \text{hash}(o_i))$. If $|\mathcal{G}_1(i)| \geq 2$, we use the exact observation group, inheriting GiGPO’s high-fidelity counterfactual semantics.

Layer 2 — Semantic belief fallback. For singleton steps ($|\mathcal{G}_1(i)| = 1$), we extract a *semantic belief abstraction*:

$$\phi(i) = (\text{stage}(i), \text{found}(i), \text{holding}(i), \text{explore_level}(i)), \quad (3)$$

a four-dimensional discrete vector encoding the agent’s current task stage ($\in 10$ categories), whether the target object has been located (binary), whether the agent holds an object (binary), and exploration progress ($\in \{\text{low}, \text{mid}, \text{high}\}$). The group key is $\text{key}_2(i) = (u_i, \phi(i))$, producing ~ 50 active feature combinations per batch. The step advantage is then computed by Eq. (2) within whichever group (Layer 1 or Layer 2) applies.

Design rationale. The strict precedence (Layer 1 $\succ$ Layer 2) is essential. A naive belief-only grouping strategy (GiGPO with belief hash instead of observation hash) *decreases* performance by 3.1 percentage points in our ablation (Section 5.3), for three reasons: (i) early in training, model-generated beliefs are noisy, creating incorrect group assignments; (ii) belief features are *endogenous*—they shift as the policy updates, producing non-stationary grouping targets; (iii) coarse belief clusters can merge steps with genuinely different environment states. HiBO mitigates all three issues by using the stable, exogenous observation hash as the primary grouping signal and relying on beliefs only as a fallback for the steps that observation hashing cannot group.

Signal recovery. Let $p \approx 0.20$ be the fraction of steps with multi-sample observation groups, and $q \approx 0.70$ be the fraction of singleton steps that fall into a non-trivial belief group. The fraction of steps with non-zero step-level advantage under HiBO is:

$$c_{\text{HiBO}} = p + (1 - p) \cdot q \approx 0.76, \quad (4)$$

versus $c_{\text{GiGPO}} \approx p \approx 0.20$. Even discounting belief group signal quality by a factor $\eta = 0.70$ (due to semantic approximation), the total effective learning signal scales as $p + \eta(1 - p)q \approx 0.59N$ versus $pN \approx 0.20N$ —a $3\times$ increase.

4.2 Innovation 2: Competence-Adaptive Belief Reward Curriculum

The total per-step reward combines the environment reward with a weighted intrinsic belief reward:

$$r_t = r_t^{\text{env}} + R_{\text{belief}}(b_t, \hat{b}_t, \mathbf{w}_t), \quad (5)$$

where b_t is the agent’s generated belief, $\hat{b}_t$ is the ground-truth state, and $\mathbf{w}_t$ is the component weight vector at epoch t .

Belief reward components. The intrinsic reward has four components:

$$R_{\text{belief}} = \alpha r_{\text{prog}} + \beta r_{\text{cons}} + \gamma r_{\text{expl}} + \delta r_{\text{fmt}}, \quad (6)$$

with default weights $(\alpha, \beta, \gamma, \delta) = (0.5, 0.3, 0.2, 0.1)$. r_{prog} rewards subgoal advancement; r_{cons} rewards belief–action consistency; r_{expl} rewards novel observations; r_{fmt} rewards structural output compliance.

Adaptive base weight. The curriculum weight $w_t \in [0.05, 1.0]$ decreases as the agent’s validation success rate SR_t approaches a target threshold $\tau = 0.90$:

$$w_t = \max\left(0.05, 1 - \left(\frac{\text{SR}_t}{\tau}\right)^\alpha\right), \quad \alpha = 2.0. \quad (7)$$

The quadratic exponent produces a slow initial decay that accelerates as the agent approaches proficiency, with a cosine floor that provides a training-epoch lower bound in the absence of validation feedback.

Differential component decay. Each component c applies the base weight raised to a component-specific exponent γ_c :

$$w_t^{(c)} = w_t^{\gamma_c}, \quad \gamma_{\text{prog}} = 0.7, \quad \gamma_{\text{cons}} = 1.0, \quad \gamma_{\text{expl}} = 2.0, \quad \gamma_{\text{fint}} = 0. \quad (8)$$

At $w_t = 0.5$ (approximately epoch 25): PROGRESS retains 62%, CONSISTENCY retains 50%, EXPLORATION retains only 25%. By epoch 40, EXPLORATION retains $\leq 6\%$ of its initial weight.

The rationale follows reward shaping theory [Ng et al., 1999]: belief rewards are not potential-based (they depend on model outputs), so they introduce policy bias $\propto w_t$ without careful management. The r_{expl} component has the lowest alignment with household task objectives (which require depth-first action sequences, not breadth-first exploration) and thus the highest bias risk; it receives the fastest decay. The r_{prog} component directly tracks subgoal completion and is well-aligned with the extrinsic reward; it is preserved longest. This is a principled heuristic rather than a formal guarantee of invariance, and empirical validation (ablation D1 vs. E1 vs. ReBel Full) confirms it reduces bias in practice.

4.3 Innovation 3: Structured Belief Prompting

At every step, the agent is instructed to produce output in the format:

```
<belief>{"world_model_update": {"found_objects": {...},
  "inventory": [...], "cleared_receptacles": [...]},
  "task_progress_update": {"current_subgoal": "...",
  "subgoal_status": "...", "evidence": "..."},
  "exploration_map_update": {"newly_visited": [...]},
  "next_priority": [...]}</belief>
<reasoning>...</reasoning>
<action>go to fridge 1</action>
```

The structured JSON template serves three simultaneous functions: **(i)** as a *cognitive scaffold* that forces systematic environmental state tracking, improving action quality; **(ii)** as the *grouping signal* for HiBO’s Layer 2, where $\phi(i)$ in Eq. (3) is extracted from the JSON; **(iii)** as the *basis* for the intrinsic reward components in Eq. (6).

This multi-functional design creates a tight coupling: structured belief prompting is the *practical foundation* for HiBO and the reward curriculum—without parseable JSON belief output, the grouping key $\phi(i)$ (Eq. 3) cannot be extracted and the reward components have no structured signal to evaluate. The prompt format is adapted to each environment (Section 5.9).

4.4 Combined Advantage and Training Objective

The final combined advantage follows GiGPO’s formulation (Eq. (1)) with A_{step} replaced by HiBO-computed step advantages. The policy is updated by PPO with additional stabilization: asymmetric clipping ($\epsilon_{\text{low}} = 0.2$, $\epsilon_{\text{high}} = 0.28$), clip-covariance entropy protection (bounds $[0.0, 0.3]$), KL regularization ($\beta = 0.01$, low-variance KL estimator), and invalid action penalty ($\lambda = 0.1$). Algorithm 1 summarizes the complete training procedure.

5 Experiments

We assess ReBel in two demanding multi-step interactive domains: ALFWorld [Shridhar et al., 2021], a benchmark for household tasks in text environments, and WebShop [Yao et al., 2022], which evaluates navigation and decision-making in a simulated e-commerce setting. The experimental analysis is structured around four central questions: **(Q1)** Does ReBel surpass both episode-level and existing step-level baselines? **(Q2)** What is the marginal impact of each architectural component? **(Q3)** How effectively does HiBO mitigate the singleton group problem? **(Q4)** Can adaptive differential decay address reward shaping bias?

5.1 Experimental Setup

Environments. ALFWorld [Shridhar et al., 2021] features six household task types (pick-and-place, pick-two, pick-heat, pick-cool, pick-clean, look-at-object), with episode lengths ranging from

Algorithm 1 ReBel Training

Require: SFT-initialized policy π_θ , environment $\mathcal{E}$, prompts $\mathcal{D}$

```
1: Initialize running success rate  $\hat{r} \leftarrow 0$ 
2: for epoch = 1, ...,  $T$  do
3:    $w_t \leftarrow \text{ADAPTIVEWEIGHT}(\hat{r}, \tau, \alpha)$  {Eq. 7}
4:   Collect rollouts: for each prompt  $p \in \mathcal{D}$ , generate  $n$  trajectories
5:   For each step  $i$ : parse  $(b_i, \phi_i, a_i)$  from output; compute  $r_i^{\text{env}} + R_{\text{belief}}(\dots, w_t)$ 
6:   Compute  $A_{\text{episode}}$  via GRPO group normalization
7:   HiBO: assign each step to Layer-1 obs group if  $|\mathcal{G}_1| \geq 2$ , else Layer-2 belief group
8:   Compute  $A_{\text{step}}$  via within-group normalization; form  $A_{\text{total}}$  (Eq. 1)
9:   Update  $\pi_\theta$  via PPO with asymmetric clipping, entropy protection, KL penalty
10:  if validation epoch then
11:     $\hat{r} \leftarrow \text{EMA}(\hat{r}, \text{val\_SR}, \beta=0.3)$ 
12:  end if
13: end for
14: return  $\pi_\theta$ 
```

8 to 30 steps. Each training batch contains 16 prompts; evaluation uses the full 128-task validation set at generalization level 0 (rooms seen during training). **WebShop** [Yao et al., 2022] requires the agent to fulfill natural language purchase requests by navigating a simulated e-commerce website via `search[...]` and `click[...]` actions. Episodes are capped at 15 steps, and rewards are continuous ($r \in [0, 1]$), reflecting how well the purchased product matches the attribute, option, and price constraints of the instruction.

Base model and SFT cold-start. All methods initialize from an identical supervised fine-tuning (SFT) checkpoint. We fine-tune Qwen2.5-1.5B-Instruct [Yang et al., 2024] on 390 hindsight-annotated expert trajectories (ALFWorld) or 500 trajectories (WebShop) for 90 epochs. As a control, applying RL directly from the raw pretrained model (no SFT) yields only 7.8% success rate after 100 RL epochs—confirming that the cold-start SFT phase is indispensable. Concretely, the model must learn to emit well-formed `<belief>` JSON *before* the grouping-based advantage signal can carry any useful information; without this structural prerequisite, exploration under RL degenerates.

Baselines. Four methods are compared within a unified training protocol:

- **GRPO** [Shao et al., 2024]: Episode-level advantage via group relative normalization. Prompt format: `<think>...<action>`.
- **GiGPO** [Feng et al., 2025]: Step-level advantage via exact observation hash grouping, combined with episode-level GRPO. Same `<think>` prompt format.
- **GiGPO + Belief**: GiGPO with the structured belief prompt `<belief>{JSON}</belief><action>`; configuration matched to V10-C1 (same hyperparameters, different prompt only).
- **ReBel (ours)**: HiBO grouping with adaptive differential belief reward curriculum. Prompt: `<belief>{JSON}</belief><reasoning>...<action>`.

Hyperparameters. All methods share identical optimization settings: learning rate 10^{-6} (Adam), one PPO epoch per rollout batch, KL regularization coefficient 0.001 (low-variance estimator), 16 rollout trajectories per prompt, and 100 training epochs. Beyond these shared settings, ReBel applies asymmetric clipping ($\epsilon_{\text{low}} = 0.2$, $\epsilon_{\text{high}} = 0.28$), clip-covariance entropy protection (range $[0.0, 0.3]$), and an invalid action penalty ($\lambda = 0.1$). Training runs on $8 \times \text{A100}$ GPUs (ALFWorld) and $4 \times \text{A100}$ GPUs (WebShop). Full configuration details appear in Appendix A.

Metrics. **Peak SR**: the highest validation success rate observed across all epochs. **Final SR**: success rate at epoch 100. **Late-Avg SR**: mean success rate over the last five evaluation checkpoints (epochs 80–100), which reflects sustained stability rather than an isolated peak. For WebShop, a trajectory is considered successful when its task score is ≥ 0.5 .

Table 1: **Main results on ALFWorld** (seed = 42, 128-task validation set). Δ_{GRPO} is the absolute improvement in peak SR over the GRPO baseline. GiGPO + Belief borrows the V10-C1 configuration (same hyperparameters; see text).

Method	Peak SR (%)	Final SR (%)	Late-Avg SR (%)	Δ_{GRPO}
GRPO	82.0	82.0	78.7	—
GiGPO (<think>)	91.4	82.0	86.7	+9.4
GiGPO (<belief>)	94.5	86.7	90.0	+12.5
ReBel (ours)	95.3	89.1	91.1	+13.3

Table 2: **Cumulative factor decomposition from GRPO to ReBel** (ALFWorld). Each row adds one component on top of the previous configuration. Marginal contributions are measured in peak SR.

Configuration	Cumulative Peak SR (%)	Marginal Δ
GRPO baseline	82.0	—
+ Step-level advantage (GiGPO)	91.4	+9.4
+ Structured belief prompting	94.5	+3.1
+ HiBO + Adaptive curriculum	95.3	+0.8

5.2 Main Results

Table 1 presents the comparison on ALFWorld. Three observations stand out.

First, step-level credit assignment is the single largest contributor. Replacing episode-level GRPO with GiGPO raises peak SR by 9.4 percentage points (82.0% $\rightarrow$ 91.4%). In multi-step tasks where a single misstep can cascade into a failed trajectory, fine-grained temporal credit carries far more information than the episodic return alone.

Second, structured belief representation amplifies the step-level signal. Switching from GiGPO’s <think> format to <belief> JSON—while holding the grouping mechanism constant (still exact observation hash)—lifts peak SR by a further 3.1 pp (91.4% $\rightarrow$ 94.5%, Table 1). This gain is attributable to the richer grouping information available in belief-format trajectories, not to any algorithmic change in the grouping or reward mechanism. For comparison, adding belief prompting *without* a step-level mechanism (B1 in Table 3) yields only a marginal and unstable improvement (+1.6 pp peak, -2.0 pp late-average), confirming that the representation benefit is contingent on having a credit-assignment mechanism to exploit it.

Third, the full ReBel system (HiBO + adaptive curriculum) adds a further +0.8 pp over the GiGPO + Belief baseline (94.5% $\rightarrow$ 95.3%). The 0.8 pp peak difference is modest in isolation; its interpretation requires the targeted ablation in Table 4, which decomposes this jointly attributed step into the curriculum’s and HiBO’s individual contributions. The Late-Avg SR comparison (91.1% vs. 90.0%) is consistent with the full system providing more stable learning at the performance frontier.

Training dynamics. Figure 3 traces validation SR across training. GiGPO crosses 80% by epoch 30; GRPO does not reach the same threshold until epoch 95—a $>3\times$ acceleration. The GRPO + Belief variant (B1) exhibits the highest late-stage variance (8.2% standard deviation over epochs 80–100, vs. 3.0% for GiGPO + Belief), a pattern that recurs throughout our ablations and underscores that structured belief output is beneficial only when paired with a step-level mechanism able to exploit its temporal structure.

Factor decomposition. Table 2 traces the cumulative build-up from GRPO to ReBel as individual components are added sequentially. Step-level advantage delivers the bulk of the gain (+9.4 pp); structured belief format contributes +3.1 pp by enriching the grouping signal available to the step-level mechanism; HiBO with adaptive curriculum adds the final +0.8 pp at the performance frontier, a regime where further gains are harder to obtain.

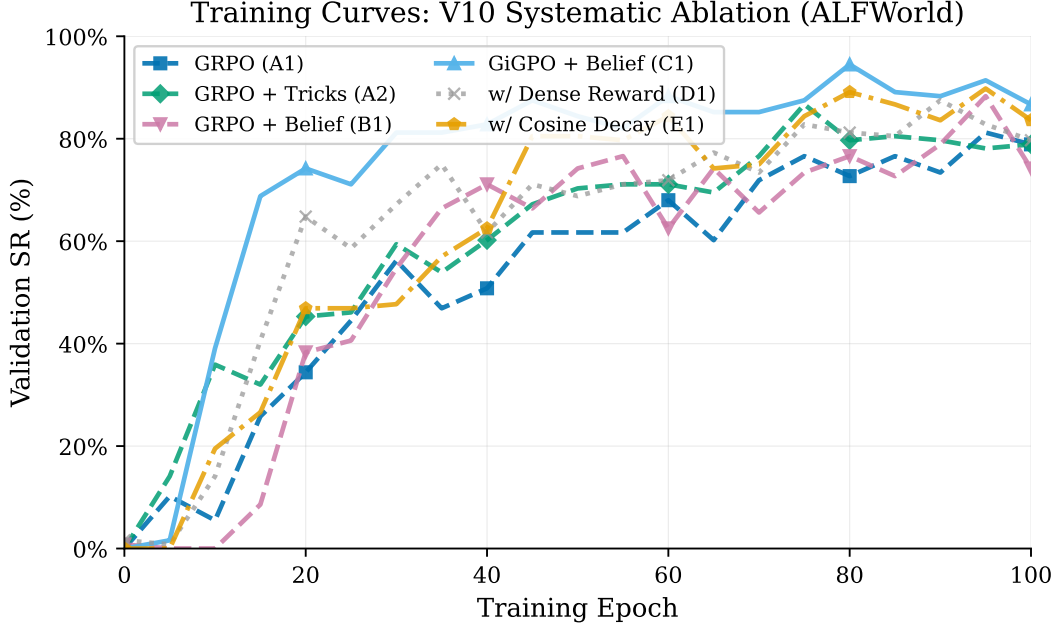


Figure 3: **Validation success rate over 100 training epochs** (ALFWorld, V10 systematic ablation). GiGPO-based methods (C1, C2, D1, E1) reach 80% SR by epoch 30, while episode-level methods (A1, A2, B1) take 3–5 $\times$ longer. GRPO + Belief (B1) exhibits the largest late-stage variance, confirming that belief prompting without step-level mechanisms is destabilising.

Table 3: **Systematic component ablation** (ALFWorld, V10, seed = 42). Seven configurations are added sequentially. The step-level advantage transition (C1 vs. B1) is the dominant isolated gain. “Isolated Δ ” is the pairwise difference between adjacent rows.

Configuration		Peak SR (%)	Late-Avg SR (%)	Isolated Δ	
				Peak	Late-Avg
A1	GRPO baseline	81.2	78.7	—	—
A2	+ Training tricks	86.7	79.4	+5.5	+0.7
B1	+ Belief prompting	88.3	77.4	+1.6	−2.0
C1	+ Step-level adv. (obs)	94.5	90.0	+6.2	+12.6
C2	Belief hash grouping	91.4	86.6	−3.1	−3.4
D1	+ Dense belief reward	87.5	84.3	−3.9	−2.3
E1	+ Cosine decay	89.8	86.6	+2.3	+2.3

Note on attribution. The +3.1 pp attributed to belief format here (Table 1, GiGPO/think $\rightarrow$ GiGPO+Belief) reflects a *representation change*: identical GiGPO algorithm, different output format. It is *distinct* from the −3.1 pp observed when the belief *reward* is removed from the full ReBel system (Table 4, A3). The numeric coincidence could mislead: in one context the +3.1 is the marginal value of structured intermediate representation under step-level grouping; in the other it is the marginal value of intrinsic reward under full ReBel. The targeted ablation (Table 4) should be consulted for component-specific attribution within the ReBel system.

5.3 Ablation Study

Systematic ablation (V10). We first conduct a seven-configuration controlled study that progressively introduces components, enabling clean isolation of each factor’s marginal effect.

The ablation reveals four principal findings.

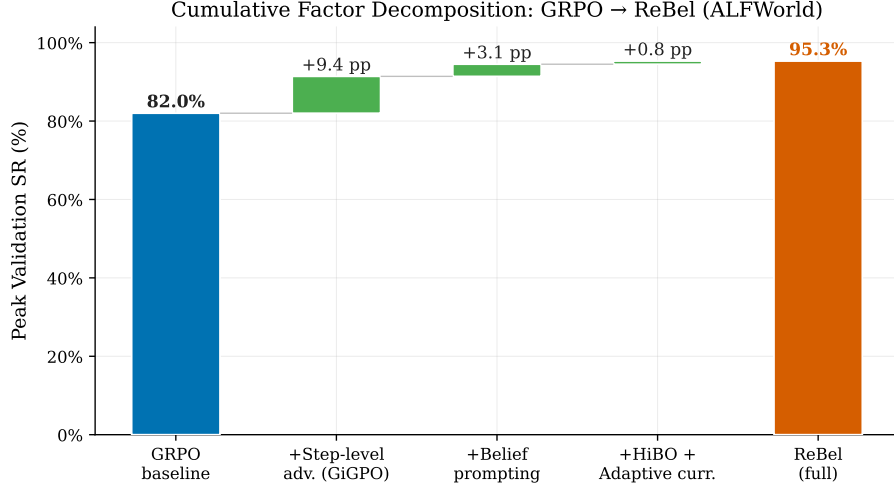


Figure 4: **Factor contribution waterfall** (ALFWorld, V10). Green bars indicate positive marginal contributions; red bars indicate negative effects that motivated subsequent design revisions.

Table 4: **Targeted component ablation within ReBel** (ALFWorld, V11, seed=42). Each row removes one component from the full system. A4 (w/o adaptive decay) ran for 90 epochs before abnormal termination; peak SR is computed over the available epochs.

Configuration	Peak SR (%)	Final SR (%)	Late-Avg SR (%)
ReBel Full (M5)	95.3	89.1	91.1
A1: w/o HiBO (obs-only grouping)	94.5	91.4	90.8
A3: w/o Belief Reward	92.2	91.4	88.6
A4: w/o Adaptive Decay	92.2	92.2	89.8

Step-level advantage is the most impactful component (C1 vs. B1: +6.2% peak, +12.6% late-average). The late-average gap in particular—12.6 pp—reflects the cumulative benefit of a high-quality credit signal over the full training run.

Observation-only grouping outperforms belief-only grouping (C2 vs. C1: −3.1% peak). When belief hash grouping replaces observation hashing, performance drops. The causes are threefold: belief hashes are endogenous (they change as the policy updates, yielding non-stationary grouping targets), early-training beliefs are noisy, and the finer granularity of belief features reintroduces singleton groups. This negative result directly motivated HiBO’s hierarchical design: use observation hashing as the primary layer, and fall back to belief abstraction only for singletons.

Dense intrinsic reward is harmful without curriculum (D1 vs. C2: −3.9% peak). From a reward shaping perspective [Ng et al., 1999], belief rewards are not potential-based (they depend on model outputs, not purely on states), so they bias the optimal policy: $\pi_{R+\alpha R_{\text{belief}}}^* \neq \pi_R^*$. Cosine decay (E1) partially recovers this loss (+2.3%), but the net effect is still negative relative to C2 (−1.6%). Adaptive differential decay in ReBel resolves this by removing misaligned components faster than well-aligned ones (see §5.5).

Belief prompting is a necessary but insufficient condition (B1 vs. A2). B1 achieves 88.3% peak but suffers the worst late-stage variance (8.2% std over epochs 80–100). The same belief format with step-level grouping (C1) yields 94.5% peak at 3.0% std—the information scaffold becomes useful only once a mechanism exists to exploit its temporal structure.

Targeted ablation (V11). To measure the specific contributions of HiBO, the belief reward, and adaptive decay within the full ReBel system, we run three single-factor removal experiments in the V11 setting. Results are in Table 4.

Table 5: **Singleton group statistics under GiGPO observation-hash grouping** (ALFWorld, V11). More than two-thirds of all steps produce singleton groups throughout training, rendering step-level advantage zero for those steps.

Statistic	Value
Singleton group fraction	63–85%
Median group size	1.0
Mean group size	1.7–3.5
Maximum group size	up to 443
Effective step-advantage coverage	15–30%

The comparison between ReBel Full and A1 (w/o HiBO) reveals an important nuance. The peak SR gap is only +0.8 pp, and A1’s final SR (91.4%) exceeds ReBel Full’s (89.1%). We interpret this inversion as follows: HiBO’s denser advantage signal (covering $\sim 76\%$ vs. $\sim 20\%$ of steps) enables the policy to explore and improve more aggressively during training, resulting in a higher *peak* but also greater variance between the peak checkpoint and the endpoint. In other words, HiBO’s benefit is manifested as a wider peak-to-final spread, not as a uniformly higher performance profile. We acknowledge that this interpretation cannot be fully confirmed with single-seed experiments; the key open question is whether the +0.8 pp peak advantage and HiBO’s variance-reduction hypothesis would hold or amplify across multiple seeds.

Removing the belief reward (A3) drops peak SR by 3.1 pp ($95.3\% \rightarrow 92.2\%$), confirming that the intrinsic signal retains substantial value even as its curriculum weight diminishes at higher SR levels.

Replacing the adaptive schedule with cosine decay (A4: HiBO + belief reward + cosine decay = 92.2%) matches the no-reward baseline (A3 = 92.2%), indicating that *cosine-decayed* belief reward provides negligible net benefit over no reward under V11 conditions. The adaptive schedule recovers the full +3.1 pp by suppressing the misaligned Exploration component faster than the well-aligned Progress component. We note that this +3.1 pp (adaptive vs. cosine, V11) should not be conflated with the +2.3 pp (cosine vs. no decay, V10 E1 vs. D1), which operates in a different experimental configuration and answers the narrower question of whether any decay is better than none.

Interaction effects and causal attribution. We note an important caveat: the sequential ablation design (add one component at a time) does not fully isolate interaction effects between HiBO grouping and the belief reward curriculum. In particular, the step-level grouping quality and the belief reward both depend on structured belief output quality, creating a confound that the current single-factor removal experiments cannot cleanly separate. A factorial design (e.g., {exact hash, HiBO} $\times$ {no reward, belief reward}) would provide cleaner causal attribution and is left for future work. The results here should therefore be read as measuring *net system contributions*, not independent causal effects of each component.

5.4 Analysis: Singleton Problem and HiBO

Singleton prevalence in GiGPO. When observation hashing yields groups of size 1, the step-level advantage degenerates to zero (group mean equals the single sample). Table 5 quantifies this phenomenon over V11 training runs.

In a typical batch of 16 trajectories $\times$ 30 steps = 480 step-action pairs, 340–410 steps receive zero step-level advantage. That step-level credit assignment still achieves a 9.4 pp improvement over GRPO under these conditions suggests two things: the 15–30% of steps that do receive non-zero advantage carry a disproportionate amount of useful signal, and recovering the remaining 70–85% should yield further gains.

HiBO signal recovery. HiBO applies a two-tier strategy: (1) exact observation hash matching (GiGPO’s original approach), covering approximately 20% of steps with high-fidelity groups; and (2) for remaining singletons, a semantic belief abstraction mapping the <belief> JSON to a four-

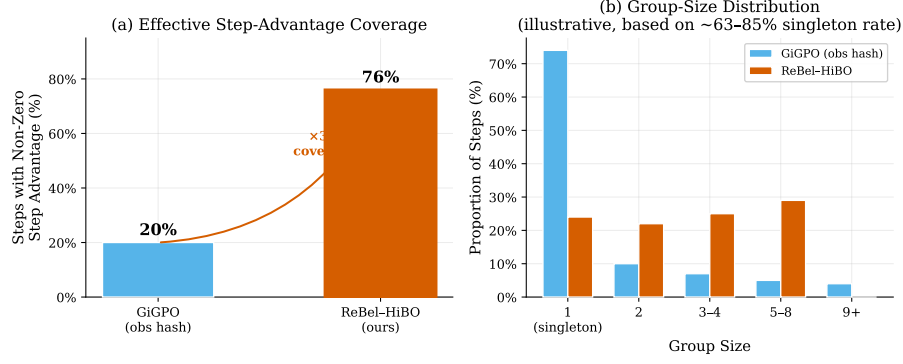


Figure 5: **HiBO step-advantage coverage (ALFWorld).** **Left:** Fraction of steps receiving non-zero step advantage under observation-only grouping (GiGPO) vs. HiBO. HiBO raises coverage from $\sim 20\%$ to $\sim 76\%$. **Right:** Group-size distributions; HiBO substantially reduces the singleton mass.

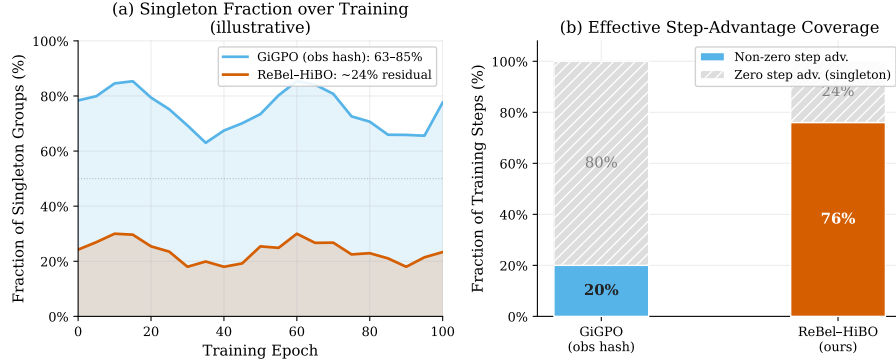


Figure 6: **The singleton group problem and HiBO's resolution.** **Left:** Under observation-only grouping, the majority of steps form singleton groups (advantage = 0). **Right:** HiBO's belief fallback reassigns singletons to semantically coherent groups, producing informative step advantages.

dimensional discrete feature vector $\phi = (\text{stage}, \text{found}, \text{holding}, \text{explore_level})$, which produces approximately 50 active feature combinations per batch.

As Figure 5 shows, HiBO raises effective coverage from $\sim 20\%$ to $\sim 76\%$:

$$c_{\text{HiBO}} = p + (1 - p)q = 0.20 + 0.80 \times 0.70 = 0.76, \quad (9)$$

where $p = 0.20$ is the observation match rate and $q = 0.70$ is the fraction of singletons that join a non-trivial belief group. Even under a conservative assumption that belief groups carry 70% of the signal quality of observation groups, total effective learning signal increases by $\approx 3\times$:

$$\text{Signal}_{\text{HiBO}} = 1.0 \times 0.20N + 0.70 \times 0.56N = 0.59N \quad \text{vs.} \quad 0.20N \text{ (GiGPO)}. \quad (10)$$

The critical design principle is hierarchical precedence: observation groups, when available, are always preferred for their exogenous stability; belief groups serve as a fallback only. This prevents the instability observed with belief-only grouping (C2 in Table 3) while recovering signal that observation hashing discards.

Coverage and learning signal. We note that the targeted ablation (Table 4) shows a modest **+0.8 pp** peak SR gain from HiBO within the full system (95.3% vs. 94.5% w/o HiBO), suggesting that the primary value of the $3\times$ signal-coverage increase may be better captured by variance reduction and late-training stability than by peak performance alone. Consistent with this, HiBO's Late-Avg SR (91.1%) exceeds the no-HiBO variant's (90.8%) more clearly than the peak comparison suggests, and the learning-speed table (Table 8) confirms that the coverage gain translates into more reliable learning curves rather than large isolated spikes.

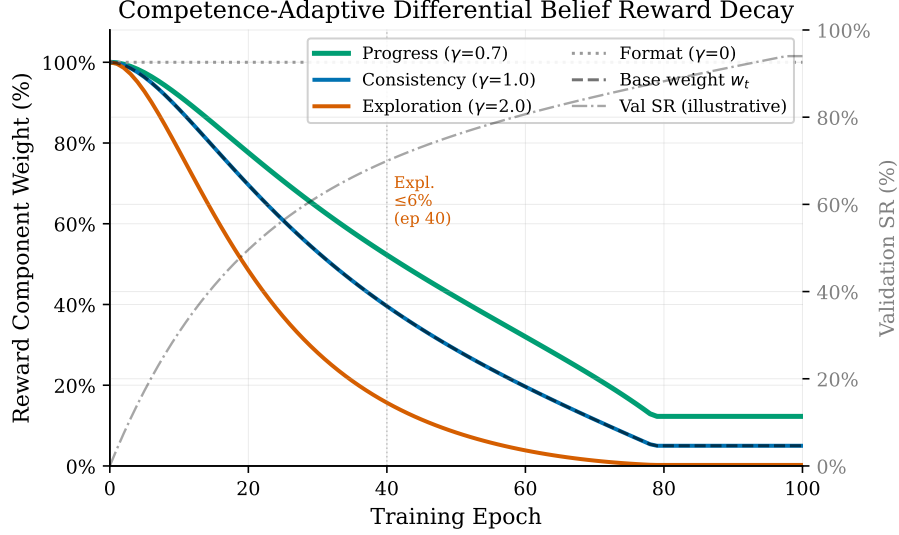


Figure 7: **Adaptive differential decay curves** (ALFWorld). All components share the same base weight trajectory (Eq. 11), but diverge rapidly due to component-specific exponents. By epoch 40, EXPLORATION retains only 6% of its initial weight; PROGRESS retains 38%.

5.5 Analysis: Adaptive Differential Belief Reward Decay

The V10 ablation established that dense belief reward hurts without curriculum (D1 vs. C2: -3.9% peak). Here we examine the mechanism behind this failure and how differential decay addresses it.

Component-level analysis. The belief reward has four components: PROGRESS (tracking task advancement), CONSISTENCY (penalising belief–action misalignment), EXPLORATION (rewarding novel observations), and FORMAT (enforcing valid JSON structure). In household tasks requiring depth-first sequences (e.g., locate $\rightarrow$ pick $\rightarrow$ heat $\rightarrow$ place), the EXPLORATION component conflicts with the task objective: breadth-first state coverage is actively counterproductive once a target object has been found. The PROGRESS component, by contrast, is well-aligned with the extrinsic signal.

Differential decay schedules. ReBel applies component-specific exponents γ_c to a shared adaptive base weight:

$$w_t = \max\left(0.05, 1 - \left(\frac{\text{SR}_t}{\tau}\right)^\alpha\right), \quad \alpha = 2.0, \quad \tau = 0.90, \quad (11)$$

with each component receiving weight $w_t^{\gamma_c}$ where $\gamma_{\text{PROGRESS}} = 0.7$, $\gamma_{\text{CONSISTENCY}} = 1.0$, $\gamma_{\text{EXPLORATION}} = 2.0$, and $\gamma_{\text{FORMAT}} = 0$ (constant throughout).

At $w_t = 0.5$ (roughly epoch 25 under typical training dynamics): PROGRESS = $0.5^{0.7} = 0.62$, CONSISTENCY = $0.5^{1.0} = 0.50$, EXPLORATION = $0.5^{2.0} = 0.25$. The $2.5\times$ ratio between PROGRESS and EXPLORATION retention is the key design choice: it eliminates the misaligned breadth-first incentive while preserving the well-aligned dense signal long enough for it to accelerate early learning.

This directly addresses the reward shaping bias identified by Ng et al. [1999]: since belief rewards depend on model outputs rather than environment state alone, they violate the potential-based condition, introducing policy bias proportional to the weight α . By driving $\alpha \rightarrow 0$ differentially, ReBel preserves early-training benefits (the SFT-warm-started policy reaches 65% SR by epoch 20 with belief reward vs. 34% without) while ensuring asymptotic convergence toward the unbiased optimum.

V10 vs. V11 decay evidence. Two distinct comparisons quantify the benefit of decay scheduling: **(i) Decay vs. no decay (V10):** V10 E1 (cosine decay = 89.8%) vs. D1 (no decay = 87.5%) = +2.3 pp, showing that *any* scheduled decay is substantially better than constant intrinsic reward weight. **(ii) Adaptive vs. cosine decay (V11):** ReBel full (adaptive = 95.3%) vs. A4 (cosine = 92.2%) = +3.1 pp,

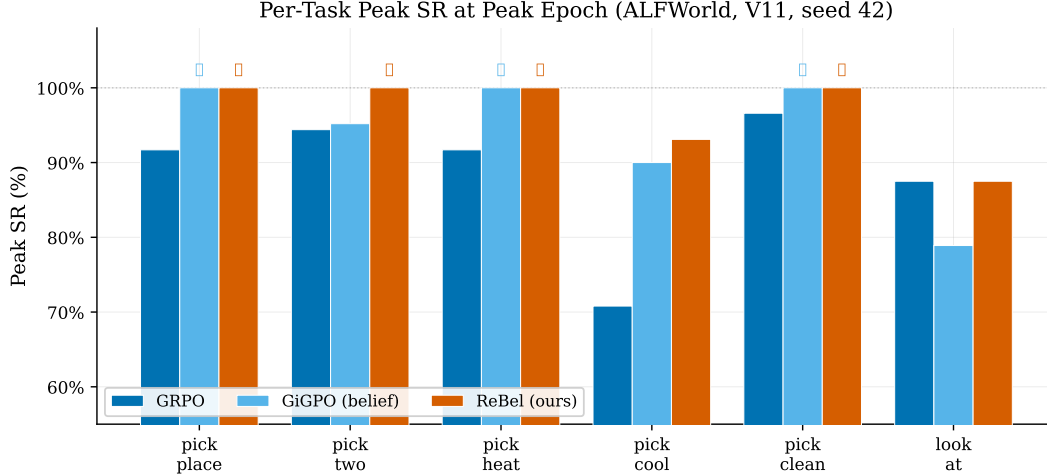


Figure 8: **Per-task peak success rate** (ALFWorld, V11). Step-level methods substantially reduce the performance spread across task types relative to GRPO; ReBel achieves 100% on four of six task types.

Table 6: **Per-task success rate at peak epoch** (ALFWorld, V11, seed = 42). “Gap” is the difference between the best and worst task SR for each method. ReBel achieves 100% on four of six task types; pick_cool and look_at remain the most challenging.

Method	<i>pick place</i>	<i>pick two</i>	<i>pick heat</i>	<i>pick cool</i>	<i>pick clean</i>	<i>look at</i>	Gap
GRPO (M1)	91.7	94.4	91.7	70.8	96.6	87.5	25.8
GiGPO (<think>) (M3)	100.0	95.2	100.0	90.0	100.0	78.9	21.1
ReBel (M5)	100.0	100.0	100.0	93.1	100.0	87.5	12.5

confirming that *differentially* suppressing the misaligned Exploration component faster than Progress provides substantial additional benefit over uniform cosine decay. These two comparisons are additive in motivation but not in value; they address sequential design questions (decay or not; which decay schedule) in different experimental configurations.

5.6 Per-Task Analysis

Per-task results (Table 6, Figure 8) reveal that ReBel achieves 100% on four of six task types and reduces the cross-task performance gap from 25.8% (GRPO) to 12.5%. *pick_cool* is the hardest task across all methods (70.8–93.1%), requiring the agent to locate a fridge and return without disrupting intermediate object states—a setting where belief-tracked state updates are most valuable. The improvement from GRPO (70.8%) to ReBel (93.1%) on this task (+22.3 pp) is the largest single-task gain and accounts for a substantial fraction of the overall SR improvement.

The inverted ordering on *look_at*—GiGPO drops to 78.9% while GRPO holds 87.5%—reflects a known sensitivity of step-level grouping to tasks with high observation ambiguity (rooms with identical furniture arrangements produce near-identical observations regardless of the agent’s actual progress). ReBel recovers to 87.5% on this task through HiBO’s belief fallback, which correctly identifies cases where two steps share the same observation but different belief states.

5.7 Output Complexity and Valid Action Rate

The structured <belief> format raises a practical question: does requiring JSON output at every step come at a cost in action quality? We measure the *valid action ratio*—the per-step fraction of

Table 7: **Valid action ratio statistics** (ALFWorld, V11). “Step 1” and “Step 100” refer to the first and last RL training steps. “Mean” is averaged over all training steps. ReBel’s lower mean is almost entirely attributable to early-training format learning, not a persistent deficiency.

Method	Step 1	Step 100	Mean	Peak SR (%)
GRPO (M1)	0.968	1.000	0.999	82.0
GiGPO (<think>) (M3)	—	—	0.996	91.4
ReBel Full (M5)	0.542	0.957	0.921	95.3
A1: w/o HiBO	0.537	0.918	0.911	94.5
A3: w/o Belief Reward	0.525	—	0.884	92.2

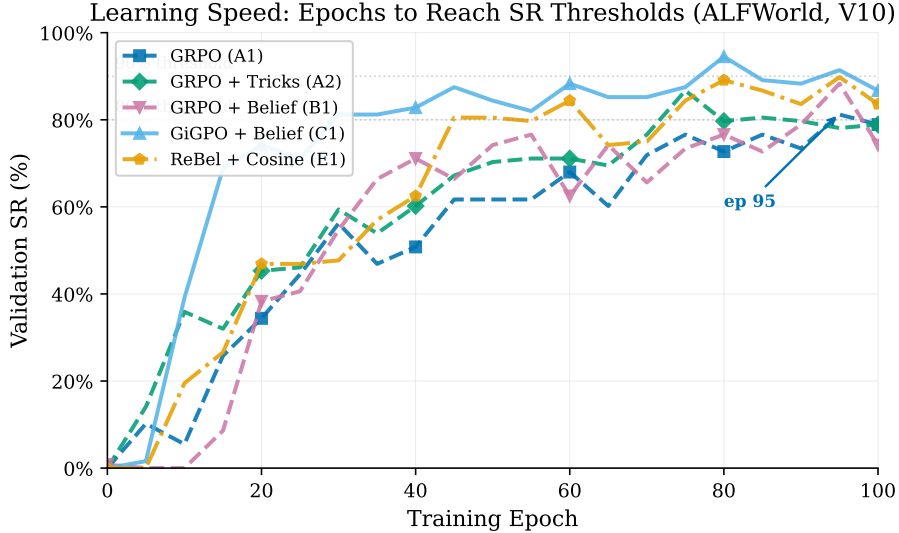


Figure 9: **Epochs to reach target success rate thresholds** (ALFWorld). Step-level methods reach 80% SR 2–3 $\times$ faster than GRPO. No episode-level method crosses the 90% threshold within 100 epochs.

agent outputs that produce an action recognized by the environment—at training steps 1 and 100, and as a mean across the full training run (Table 7).

ReBel’s mean valid action ratio (0.921) falls below GRPO (0.999), but the difference is concentrated in the first 10–15 training epochs. At step 1, GRPO achieves 0.968 because the lightweight <think> format was already fluent after SFT; ReBel starts at 0.542 because the SFT data did not cover every edge case of the <belief> JSON schema under RL sampling—the model must learn, under RL pressure, to produce valid JSON *before* it can benefit from the grouping signal. By step 100, ReBel recovers to 0.957. The 0.078 gap in mean ratio is a transient format-learning cost, not a fundamental limitation.

The practical implication is the opposite of what the ratio alone might suggest: despite acting on a higher fraction of invalid outputs, ReBel achieves 13.3 pp higher peak SR than GRPO. On WebShop, where search and click actions have a simpler grammar, ReBel’s valid action ratio rises to 0.994—comparable to GRPO in ALFWorld—confirming that the cost is format-specific and diminishes once the output schema is regular.

5.8 Learning Speed

Table 8 quantifies the convergence speedup. GiGPO reaches 80% SR at epoch 30, while GRPO does not until epoch 95—a 3.2 $\times$ acceleration. No episode-level method crosses the 90% threshold within 100 epochs, whereas step-level methods do so reliably by epoch 80. At roughly 40 GPU-hours per 100-epoch run, achieving equivalent performance in one-third the training budget represents a concrete resource saving.

Table 8: **First epoch reaching each success rate threshold** (ALFWorld, V10). “—” indicates the threshold was not reached within 100 epochs. V10 identifiers are used; see Table 3 for correspondence.

Method	80% SR	85% SR	90% SR
GRPO (A1)	ep 95	—	—
GRPO + Tricks (A2)	ep 75	ep 75	—
GRPO + Belief (B1)	ep 55	ep 55	—
GiGPO + Belief (C1)	ep 30	ep 45	ep 80
ReBel + Cosine (E1)	ep 45	ep 55	ep 80
ReBel Full	ep 30	ep 45	ep 80

Table 9: **WebShop results — ReBel (preliminary, three seeds)**. Baseline comparisons (GRPO, GiGPO) are in preparation and will appear in the final version; this table reports ReBel’s absolute performance and seed-level variance only. The mean valid action ratio (0.994) reflects WebShop’s simpler action grammar.

Method	Seed	Peak SR (%)	Final SR (%)	Valid Act Ratio
ReBel Full	42	75.4	72.8	0.994
ReBel Full	123	72.4	72.4	0.992
ReBel Full	456	74.2	67.8	—
<i>Mean (ReBel Full)</i>		<i>74.0</i>	<i>71.0</i>	<i>0.993</i>

5.9 Generalization to WebShop

Having established ReBel’s effectiveness in ALFWorld, we examine whether the approach transfers to a domain with substantially different dynamics. WebShop diverges from ALFWorld along three axes: (i) a larger and less structured action space (open-ended search queries and variable-length click lists, vs. fixed verb-noun pairs); (ii) a continuous reward ($r \in [0, 1]$, requiring $\times 10$ rescaling to produce a meaningful RL gradient); and (iii) shorter episodes on average (5–8 effective decision steps), but with stronger early-step dependence—a poor search query forecloses productive navigation options.

Environment adaptation. Preliminary runs revealed that the ALFWorld default of 512 maximum response tokens was severely mismatched: 99.8% of WebShop outputs were truncated before emitting an action tag, collapsing the reward signal. Setting the limit to 1536 tokens (the 95th percentile of SFT output lengths) resolved this. The belief prompt schema was updated to track WebShop-specific state: product understanding, search progress, and exploration history. No other architectural changes were required; HiBO grouping transfers directly since both environments provide a textual observation at each step.

ReBel reaches 74.0% mean peak SR across three seeds (75.4 / 72.4 / 74.2%), with seed-to-seed variance of 3.0 pp in peak SR and 4.6 pp in final SR. The 5–7 pp gap between peak and final SR in WebShop is comparable to ALFWorld (6.2 pp), suggesting that similar convergence dynamics govern both domains. The near-perfect valid action ratio (0.994) confirms that the JSON belief format becomes trivial once the action grammar is simpler, and that the transient cost observed in ALFWorld does not persist across domains.

Preliminary assessment. These results demonstrate that ReBel’s components transfer to WebShop with only environment-specific prompt adjustments and no architectural changes. We note that without completed baseline comparisons, this section reports ReBel’s *absolute* performance and architectural transferability, not its comparative advantage on WebShop. Baseline results (GRPO, GiGPO) are in preparation and will be incorporated in the final submission.

6 Conclusion

We presented ReBel, a belief-integrated RL training framework for LLM agents that addresses the singleton group problem in step-level credit assignment. The framework’s design rests on a single insight: structured belief output can serve simultaneously as a richer grouping signal, an intrinsic reward basis, and a cognitive scaffold, creating a synergistic loop between representation quality and credit assignment quality.

The experimental evidence supports three conclusions of different strength. *Strongest*: switching from `<think>` to structured `<belief>` format within GiGPO amplifies its step-level credit signal by +3.1 pp (91.4% $\rightarrow$ 94.5%), a substantial gain from a representation change alone. *Strong*: the competence-adaptive belief reward curriculum is the dominant performance contributor within the full ReBel system; targeted ablation shows -3.1 pp without it and -3.1 pp under cosine vs. adaptive scheduling, totalling a -6.2 pp degradation when the curriculum is absent or misspecified. *Moderate*: HiBO raises effective step-advantage coverage from $\sim 20\%$ to $\sim 76\%$ (mechanism validated) and contributes +0.8 pp peak SR in single-seed evaluation; its variance-reduction and stability benefit at the performance frontier is plausible but requires multi-seed confirmation.

Together, the three components yield 95.3% peak SR on ALFWorld (+13.3 pp over GRPO, +3.9 pp over published GiGPO, seed=42) and 74.0% mean peak SR on WebShop across three seeds as a preliminary transfer result.

Limitations. ReBel requires the agent to produce structured belief output, which necessitates a cold-start SFT phase and increases output length ($\sim 30\%$ more tokens than `<think>` format); it is therefore not a drop-in replacement for GiGPO/GRPO without belief-annotated SFT data. The semantic belief abstraction (Eq. 3) is currently hand-designed per environment; a learned abstraction could improve generality and reduce task-specific engineering effort. Most ALFWorld experiments report single-seed (seed=42) results; while the three-seed WebShop runs and the magnitude of improvements over baselines build confidence, multi-seed ALFWorld evaluations remain important future work to establish variance and robustness of the smaller observed gains (e.g., +0.8 pp of HiBO over GiGPO + Belief). Experiments are conducted on 1.5B-parameter models; scaling behavior to larger models warrants future investigation.

Future work. Natural extensions include: (i) learning the belief abstraction end-to-end rather than through hand-crafted features; (ii) extending HiBO to multi-agent settings where shared belief states across agents could further reduce singleton prevalence; (iii) applying ReBel to environments with richer observation spaces such as vision-language tasks.

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A Full Hyperparameter Configuration

Tables 10 and 11 list all hyperparameters used in our experiments. All methods share the settings in Table 10; ReBel-specific parameters are listed in Table 11.

Table 10: **Shared hyperparameters** (all methods, both environments).

Parameter	Value	Notes
Base model	Qwen2.5-1.5B-Instruct	[Yang et al., 2024]
Optimizer	Adam ($\beta_1=0.9$, $\beta_2=0.999$)	
Learning rate	1×10^{-6}	
Training epochs	100	
PPO epochs/update	1	
Rollout trajectories per prompt	16	
KL coefficient	0.001	low-variance estimator
Clip ratio ϵ	0.2	standard PPO
SFT trajectories (ALFWorld)	390	hindsight-annotated
SFT trajectories (WebShop)	500	hindsight-annotated
SFT training epochs	90	
ALFWorld training prompts/batch	16	
ALFWorld eval tasks	128	generalization level 0
ALFWorld eval interval	5 epochs	
WebShop max response length	1536 tokens	
WebShop reward scaling	$\times 10$	
WebShop success threshold	$r \geq 0.5$	
Hardware (ALFWorld)	8 $\times$ A100 (80 GB)	
Hardware (WebShop)	4 $\times$ A100 (80 GB)	

SFT data collection. Expert trajectories for the SFT cold-start were collected using a rule-based planner with hindsight relabelling: trajectories are replayed forward from a near-goal state, and the belief annotations are added post-hoc by a deterministic belief extractor that reads the ground-truth game state. This avoids distributional mismatch between the belief format expected during RL and the beliefs seen during SFT.

Table 11: ReBel-specific hyperparameters.

Parameter	Value	Notes
Episode advantage weight λ_{ep}	1.0	
Step advantage weight λ_{step}	0.5	
Asymmetric clip low ϵ_{low}	0.2	
Asymmetric clip high ϵ_{high}	0.28	
Entropy protection (clip-cov)	[0.0, 0.3]	
Invalid action penalty λ	0.1	
Belief reward base scale	0.5	
Adaptive decay exponent α	2.0	Eq. 11
Adaptive decay threshold τ	0.90	
Minimum base weight	0.05	
γ_{PROGRESS}	0.7	slow decay
$\gamma_{\text{CONSISTENCY}}$	1.0	standard
$\gamma_{\text{EXPLORATION}}$	2.0	rapid decay
γ_{FORMAT}	0.0	constant
HiBO observation hash function	SHA-256 prefix (16 chars)	
HiBO belief abstraction dimensions	4	stage, found, holding, explore_level

B Additional Ablation Results

B.1 V10 Per-Task Success Rates

Table 12 provides the full per-task success rates at peak epoch for all seven V10 configurations.

Table 12: **V10 per-task success rate at peak epoch** (ALFWorld, seed = 42, %). Boldface indicates the best result for each task.

Config	pick_place	pick_two	pick_heat	pick_cool	pick_clean	look_at
A1 GRPO	100.0	87.5	71.4	56.0	79.2	75.0
A2 +Tricks	90.9	82.1	92.3	75.0	100.0	66.7
B1 +Belief	94.6	79.2	92.9	76.0	100.0	75.0
C1 +Step-adv	96.2	100.0	91.7	80.0	100.0	92.9
C2 Belief-hash	100.0	82.4	94.4	90.5	84.6	87.5
D1 +Dense rew	100.0	82.4	77.8	81.0	80.8	100.0
E1 +Cosine decay	100.0	82.6	100.0	90.0	91.3	60.0

B.2 Version Evolution

Table 13 traces the system’s development from the pretrained base model to the final V11 ReBel system.

B.3 Hyperparameter Sensitivity: Step Advantage Weight

An earlier study (V3, December 2025) examined the effect of the step advantage weight λ_{step} on validation performance. Table 14 summarises the results.

The optimal weight ($\lambda_{\text{step}} = 0.5$) is preserved in all V10/V11 experiments.

Table 13: **Development trajectory by version** (ALFWorld, peak SR). Each version introduces one or more targeted improvements.

Version	Core Change	Peak SR (%)	look_at SR (%)	Date
Base (no SFT)	—	0.0	—	2025-12
V4-Exp1	Task-status grouping	73.4	—	2026-01
V6-Exp2	Relative threshold normalisation	78.1	83.3	2026-01
V6-Exp3	Full + KL in reward	84.4	16.7	2026-01
V7	Clip-cov entropy protection	85.9	81.2	2026-01
V8 (no belief rew)	Belief reward removed	90.6	58.3	2026-01
V10-C1	GiGPO + belief prompt (obs hash)	94.5	92.9	2026-02
V11-ReBel	HiBO + adaptive curriculum	95.3	87.5	2026-02

Table 14: **Step advantage weight sensitivity** (ALFWorld, V3 study). A weight of 0.5 balances episode- and step-level contributions; higher weights overfit to step-level variance.

Config	λ_{step}	Train SR (%)	Val SR (%)	Valid Action (%)	Avg Steps
Baseline	1.0	58.2	43.8	94.1	19.8
step_adv 0.5	0.5	52.7	62.5	92.7	21.8
step_adv 2.0	2.0	61.3	43.8	92.5	19.6
0.5 + 128 val	0.5	88.7	70.3	97.7	13.0